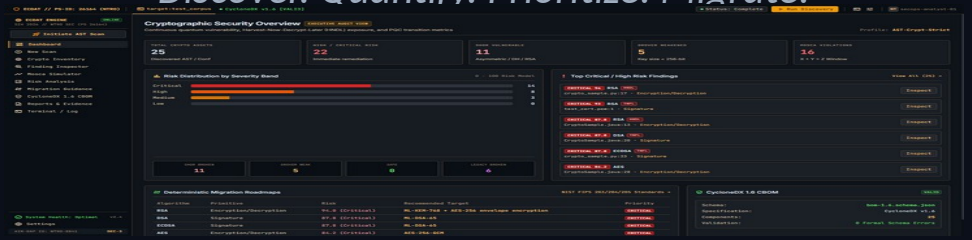
QuantumShield

Cryptographic Risk Intelligence & Post-Quantum Readiness Platform

Discover. Quantify. Prioritize. Migrate.



Team Diet Claude | MG Solutions Build Spark Hackathon

https://quantumshield-seven.vercel.app

Your organization cannot migrate cryptography it cannot see.

Hidden Asset

Source Code

Hidden Asset

Dependencies

Hidden Asset

X.509 Certificates

Hidden Asset

Keystores

Hidden Asset

Server Configurations

Hidden Asset

JVM Bytecode

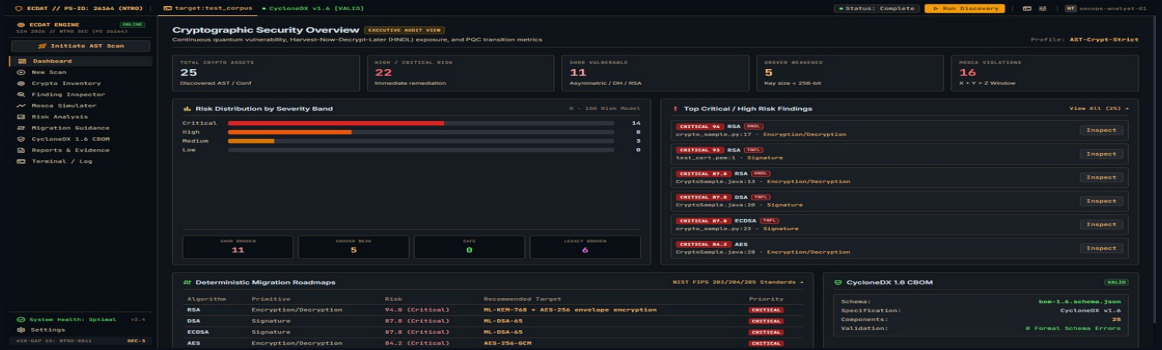
The Quantum Consequences:

Shor: Breaks RSA/ECC/DH | Grover: Halves Symmetric Security | HNDL: Retroactive Decryption | TNFL: Retroactive Forgery

QuantumShield turns invisible cryptography into actionable risk intelligence.

DISCOVER | INVENTORY | ANALYZE | PRIORITIZE | REMEDIATE | MONITOR

Offline-first | Evidence-backed | Standards-native | Migration-oriented



A six-stage cryptographic intelligence pipeline

01 Discovery

Source, deps, certs, keystores, configs, bytecode

02 Evidence

Normalization: File + line + snippet + confidence

03 Deduplication

Normalized path + line + algorithm + usage

04 Classification

Shor-broken / Grover-weakened / safe

05 Risk Modeling

Mosca inequality + 5-factor risk score

06 Remediation

PQC recommendations + CBOM + reports

From algorithm detection to measurable business risk

Mosca's Inequality: X + Y > Z

X: Migration Time | Y: Security Lifetime | Z: CRQC Horizon

5-Factor Risk Model

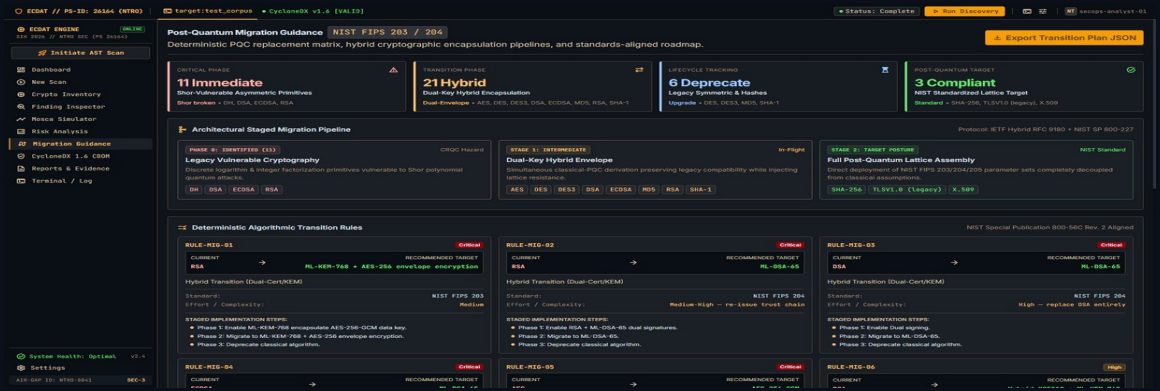
Quantum Exposure (35%)
Business Criticality (25%)
Exposure Surface (15%)
Data Sensitivity (15%)
Inversed Agility (10%)

Output: 0-100 Risk Score (Critical / High / Medium / Low)

From finding → recommendation → migration plan

Vulnerable Classical ——— ML-KEM / ML-DSA / SLH-DSA

NIST FIPS 203 / 204 / 205 Standards Mapping



Turn cryptographic findings into portable security evidence

Export Format

CycloneDX 1.6 CBOM

Export Format

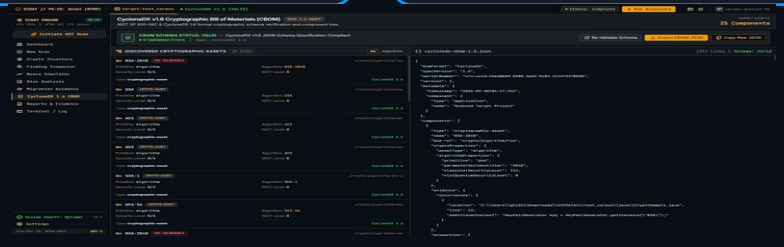
SARIF 2.1.0

Export Format

JSON / CSV

Export Format

HTML / PDF / Markdown



Built and validated as a real working system

71/71	1.000	0.974	0 Errors
Automated Tests Passed	Precision (Controlled Corpus)	Recall (Controlled Corpus)	CBOM Schema Validation

External Targets: OpenSSL 3.3.0, CPython 3.12.3, OpenSSH 9.7

Built for the gap between vulnerability scanning and quantum readiness

Generic SAST/SCA

Focus: CVEs and Library Versions
No Quantum Classification
No HNDL/TNFL Modeling
No CBOM standards

QuantumShield

Cryptographic Visibility
Shor/Grover Classification
HNDL/TNFL Analysis
Mosca Simulation
Deterministic Risk Scoring
NIST PQC Recommendations
CBOM Generation
Offline-First Analysis

Make post-quantum readiness measurable

SECURITY

Know where vulnerable cryptography exists.

RISK

Understand which assets require attention first.

MIGRATION

Create evidence-backed PQC migration priorities.

Roadmap (Planned):

Continuous Monitoring → Migration Tracking → Expanded Language Coverage → Enterprise Integration

QuantumShield | Discover. Quantify. Prioritize. Migrate.

https://quantumshield-seven.vercel.app | GitHub: /shaikmohammedyasin-create/QuantumShield